SREE 2024: Science example

Martyna Citkowicz

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```r
library(metaselection)
library(metadat)
library(psymetadata)
library(metafor)
```

```
## Loading required package: Matrix

## Loading required package: numDeriv

##
## Loading the 'metafor' package (version 4.8-0). For an
## introduction to the package please type: help(metafor)
```

```r
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.2.0     v readr     2.2.0
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.2     v tibble    3.3.1
## v lubridate 1.9.5     v tidyr     1.3.2
## v purrr     1.2.1

## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x tidyr::expand() masks Matrix::expand()
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## x tidyr::pack()   masks Matrix::pack()
## x tidyr::unpack() masks Matrix::unpack()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(here)
```

```
## here() starts at C:/GitHub/metaselection
```

```r
devtools::load_all()
```

```
## i Loading metaselection
```

Science Meta-analysis

```r
sci <- read_csv(here('tests/testdata','ScienceMeta.csv'))
```

```
## Rows: 292 Columns: 40
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (11): es_id, short_cite, long_cite, intervention_delivery, intervention_...
## dbl (29): study_id, yi, vi, sci_hs_bio, sci_hs_chem, sci_hs_phys, sci_k8_lif...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
sci <- sci |>
  rename(esid = es_id, studyid = study_id) |>
  mutate(
    d = yi,
    Va = vi,
    Var_d = Va,
    sd_d = sqrt(Var_d),
    var_d = Var_d,
    p_onesided = 1 - pnorm(d / sd_d)
  )

length(unique(sci$studyid)); length(unique(sci$esid))
```

```
## [1] 96
```

```
## [1] 292
```

Mean Only

Unadjusted estimate: mean only

```r
sci_unadj_mean <- rma.uni(
  data = sci,
  yi = d,
  sei = sd_d,
  method = "ML"
)
sci_unadj_mean
```

```
##
## Random-Effects Model (k = 292; tau^2 estimator: ML)
##
## tau^2 (estimated amount of total heterogeneity): 0.1421 (SE = 0.0189)
## tau (square root of estimated tau^2 value):      0.3770
## I^2 (total heterogeneity / total variability):   68.18%
```

```
## H^2 (total variability / sampling variability):  3.14
##
## Test for Heterogeneity:
## Q(df = 291) = 915.8432, p-val < .0001
##
## Model Results:
##
## estimate      se     zval    pval   ci.lb   ci.ub
##   0.4402  0.0285  15.4445  <.0001  0.3844  0.4961  ***
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

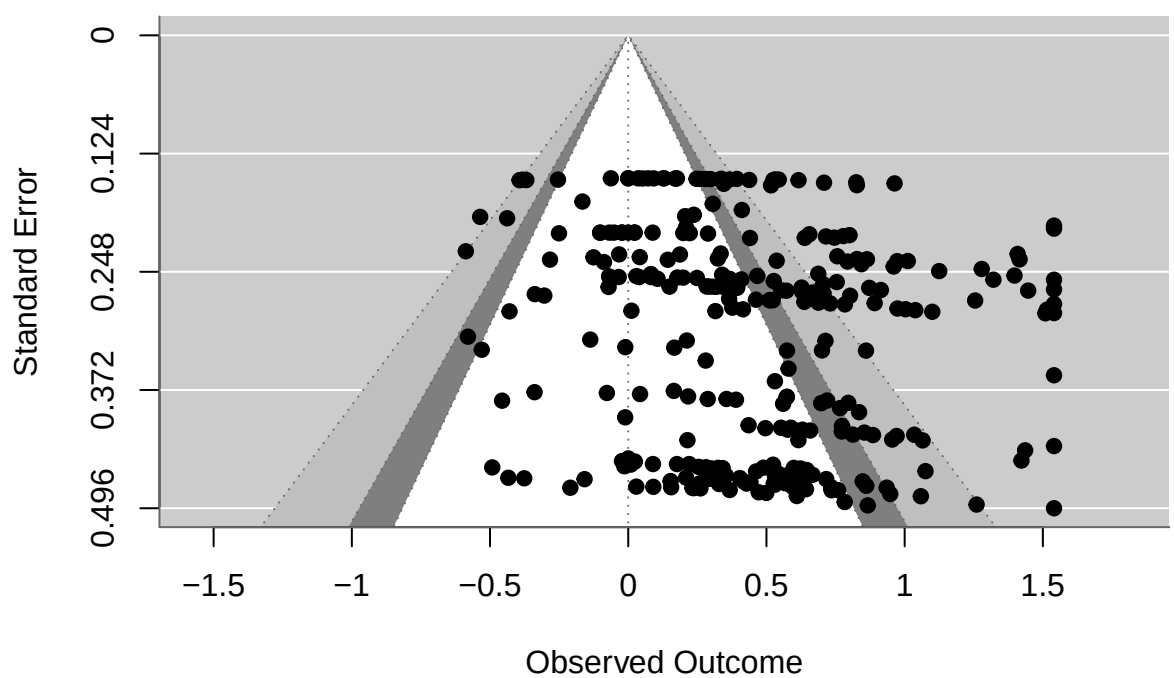
```
sci_unadj_mean_cil <- sci_unadj_mean$tau2 - 1.96*sci_unadj_mean$se.tau2
sci_unadj_mean_ciu <- sci_unadj_mean$tau2 + 1.96*sci_unadj_mean$se.tau2
sci_unadj_mean_cil; sci_unadj_mean_ciu
```

```
## [1] 0.1050343
```

```
## [1] 0.1792096
```

Forest plot

```
funnel(sci_unadj_mean, refline = 0, level = c(90, 95, 99))
```



CHE estimate: mean only

```
V_mat <- vcalc(
    vi = vi,
    cluster = studyid,
    obs = esid,
    data = sci,
    rho = 0.5,
    sparse = TRUE
  )

sci_che <- rma.mv(yi = yi,
                  V = V_mat,
                  random = ~ 1 | studyid / esid,
                  data = sci,
                  sparse = TRUE,
                  digits = 3) |>
    robust(cluster = studyid, clubSandwich = TRUE)

sci_che
```

```
##
## Multivariate Meta-Analysis Model (k = 292; method: REML)
##
## Variance Components:
##
##            estim   sqrt  nlvls  fixed        factor
## sigma^2.1  0.171  0.413     96     no       studyid
## sigma^2.2  0.037  0.192    292     no  studyid/esid
##
## Test for Heterogeneity:
## Q(df = 291) = 853.675, p-val < .001
##
## Number of estimates:   292
## Number of clusters:    96
## Estimates per cluster: 1-39 (mean: 3.04, median: 1)
##
## Model Results:
##
## estimate     se¹   tval¹     df¹   pval¹  ci.lb¹  ci.ub¹
##    0.453  0.053   8.587   90.95   <.001   0.349   0.558   ***
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## 1) results based on cluster-robust inference (var-cov estimator: CR2,
##    approx t-test and confidence interval, df: Satterthwaite approx)
```

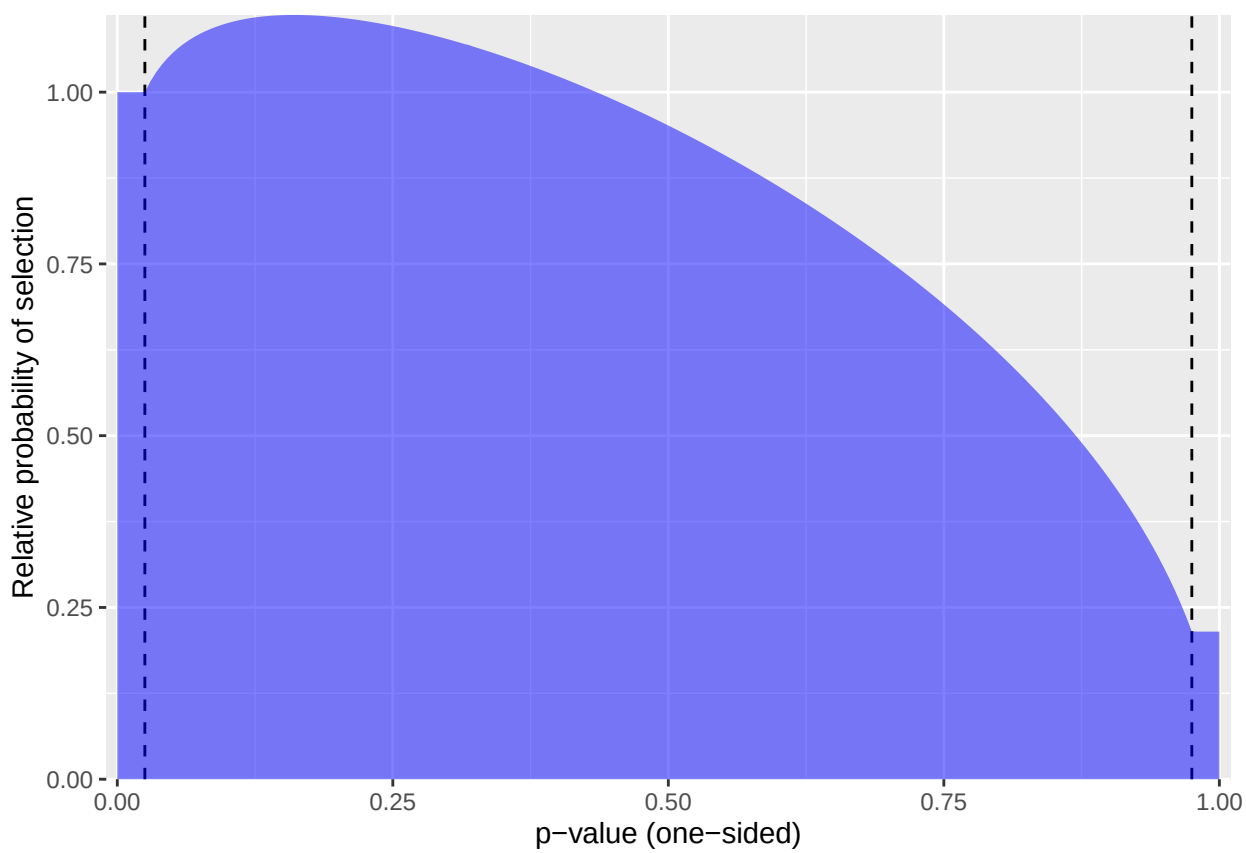
Adjusted estimate in metaselection: Beta model mean only

```r
sci_adj_ms_beta_mean <- selection_model(
  dat = sci,
  yi = d,
  sei = sd_d,
  pi = p_onesided,
  cluster = studyid,
  selection_type = "beta",
  steps = c(.025,.975),
  make_sandwich = TRUE
)
sci_adj_ms_beta_mean
```

```
##    param   Est     SE p_value CI_lo CI_hi
##     beta 0.340 0.1139 0.00279 0.117 0.564
##     tau2 0.204 0.0329      NA 0.148 0.279
##  lambda1 1.099 0.1584 0.51239 0.829 1.458
##  lambda2 1.519 0.2555 0.01296 1.092 2.112
```

Selection plot: Beta model mean only

```r
selection_plot(sci_adj_ms_beta_mean)
```



Adjusted estimate in metaselection: Step 3PSM model mean only

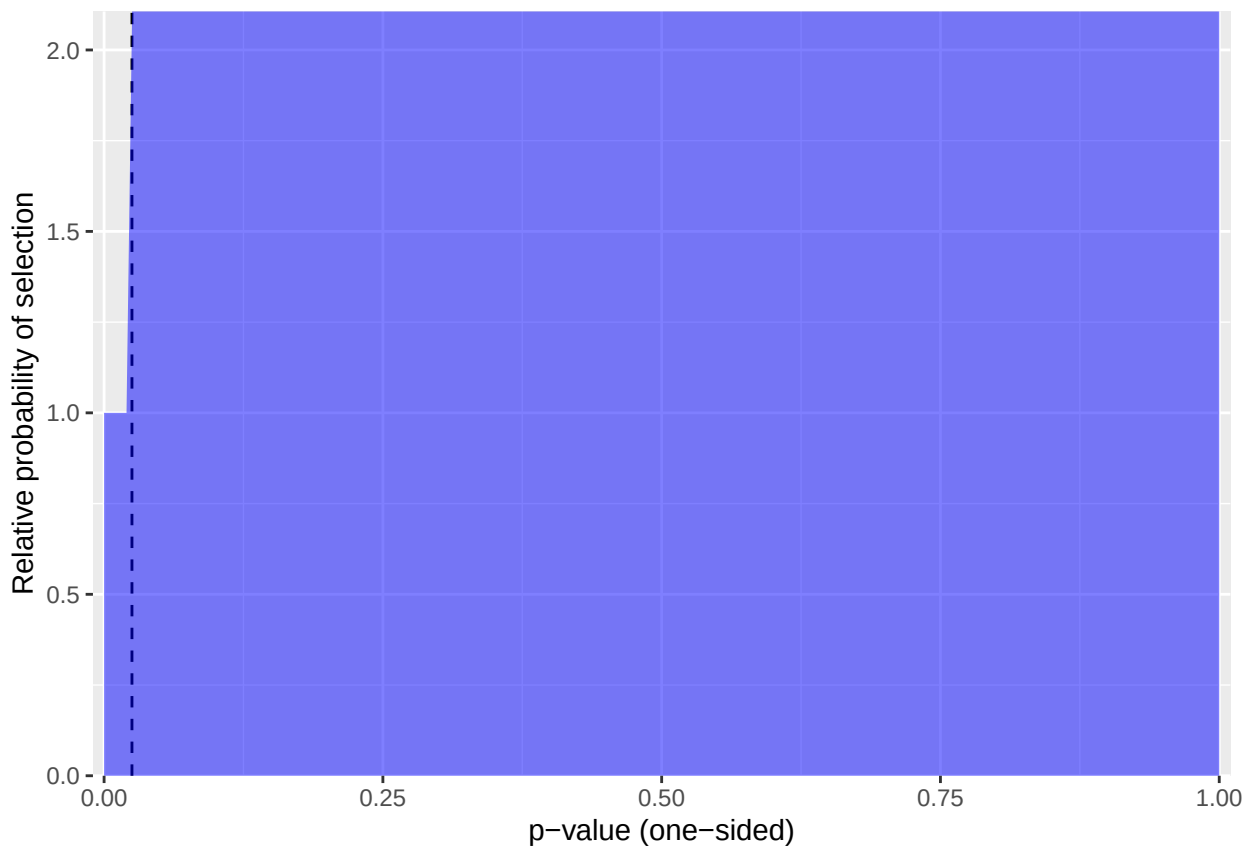
```
sci_adj_ms_step3_mean <- selection_model(
  dat = sci,
  yi = d,
  sei = sd_d,
  cluster = studyid,
  selection_type = "step",
  steps = 0.025
)

sci_adj_ms_step3_mean
```

```
##    param   Est     SE  p_value CI_lo CI_hi
##     beta 0.572 0.0620 2.60e-20 0.451 0.694
##     tau2 0.155 0.0324       NA 0.103 0.234
##  lambda1 2.106 0.5095 2.08e-03 1.311 3.384
```

Selection plot: Step 3 PSM model mean only

```
selection_plot(sci_adj_ms_step3_mean)
```



Adjusted estimate in metaselection: Step 4PSM model mean only

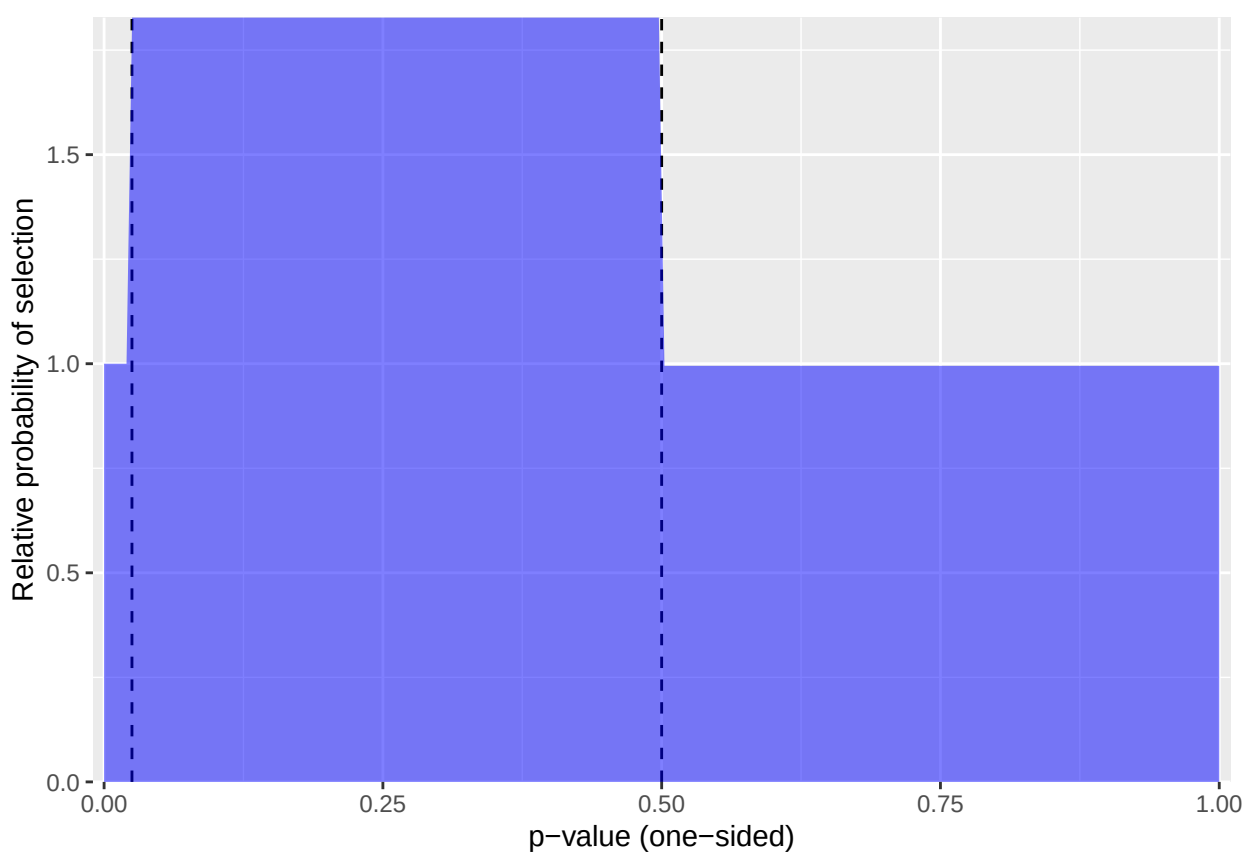
```
sci_adj_ms_step4_mean <- selection_model(
  dat = sci,
  yi = d,
  sei = sd_d,
  cluster = studyid,
  selection_type = "step",
  steps = c(0.025, .5)
)

sci_adj_ms_step4_mean
```

```
##    param   Est     SE  p_value CI_lo CI_hi
##     beta 0.478 0.0678 1.77e-12 0.345 0.611
##     tau2 0.185 0.0345       NA 0.128 0.266
##  lambda1 1.828 0.4197 8.63e-03 1.165 2.866
##  lambda2 0.995 0.3619 9.89e-01 0.488 2.029
```

Selection plot: Step 4PSM model mean only

```
selection_plot(sci_adj_ms_step4_mean)
```



Moderator

Unadjusted estimate: moderator

```
sci_unadj_mod <- rma.uni(
  data = sci,
  yi = d,
  sei = sd_d,
  method = "ML",
  mods = ~ outcome_type_author
)
sci_unadj_mod
```

```
##
## Mixed-Effects Model (k = 292; tau^2 estimator: ML)
##
## tau^2 (estimated amount of residual heterogeneity):     0.1421 (SE = 0.0189)
## tau (square root of estimated tau^2 value):             0.3770
## I^2 (residual heterogeneity / unaccounted variability): 68.13%
## H^2 (unaccounted variability / sampling variability):   3.14
## R^2 (amount of heterogeneity accounted for):            0.01%
##
## Test for Residual Heterogeneity:
## QE(df = 290) = 915.4359, p-val < .0001
##
## Test of Moderators (coefficient 2):
## QM(df = 1) = 1.8224, p-val = 0.1770
##
## Model Results:
##
##                      estimate      se    zval    pval    ci.lb   ci.ub
## intrcpt                0.3758  0.0556  6.7637  <.0001   0.2669  0.4848  ***
## outcome_type_author    0.0874  0.0647  1.3500  0.1770  -0.0395  0.2143
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
sci_unadj_mod_cil <- sci_unadj_mod$tau2 - 1.96*sci_unadj_mod$se.tau2
sci_unadj_mod_ciu <- sci_unadj_mod$tau2 + 1.96*sci_unadj_mod$se.tau2
sci_unadj_mod_cil; sci_unadj_mod_ciu
```

```
## [1] 0.1050231
```

```
## [1] 0.1791935
```

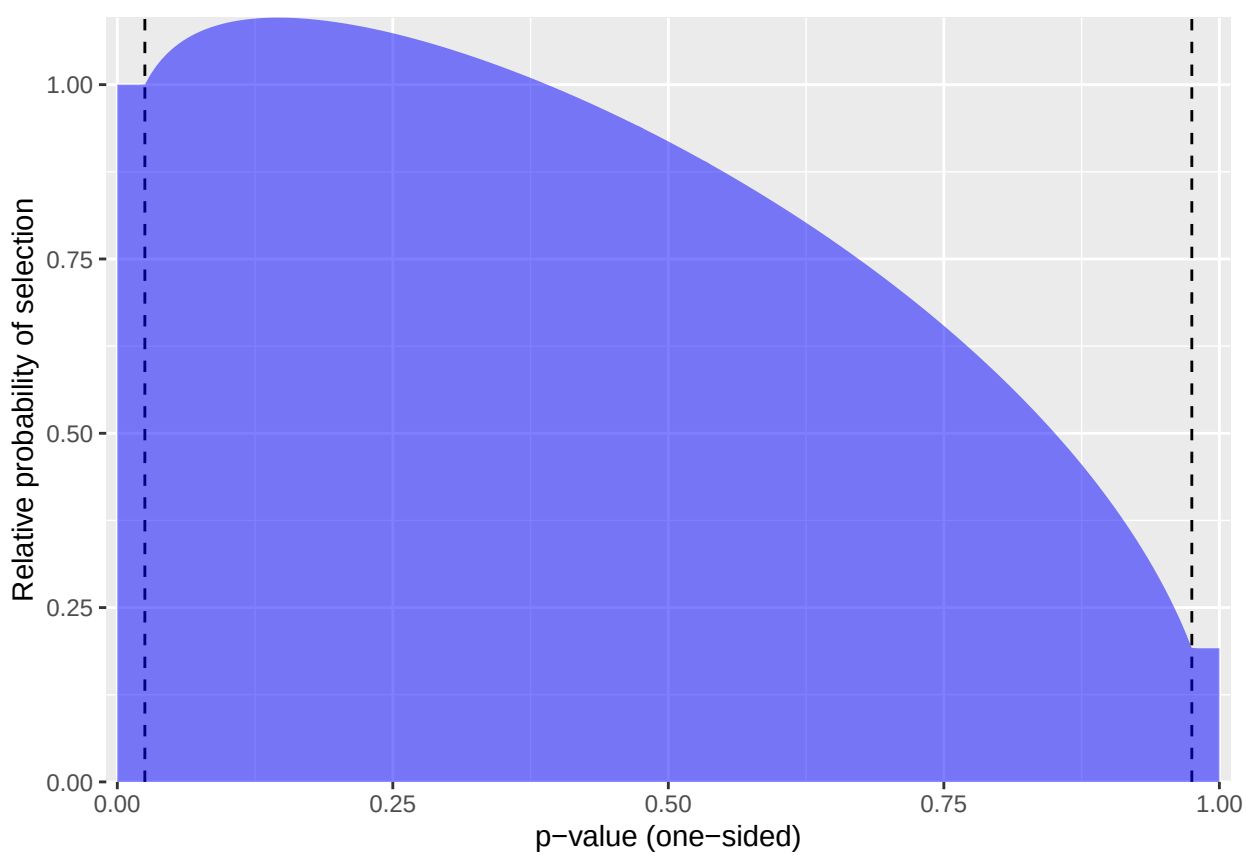
Adjusted estimate in metaselection: Beta model moderator

```r
sci_adj_ms_beta_mod <- selection_model(
  dat = sci,
  yi = d,
  sei = sd_d,
  pi = p_onesided,
  cluster = studyid,
  selection_type = "beta",
  steps = c(.025,.975),
  make_sandwich = TRUE,
  mean_mods = ~ outcome_type_author
)
sci_adj_ms_beta_mod
```

```
##                      param   Est     SE p_value  CI_lo CI_hi
##           beta_(Intercept) 0.226 0.2287   0.323 -0.222 0.674
##  beta_outcome_type_author 0.136 0.1755   0.438 -0.208 0.480
##                      tau2 0.208 0.0331      NA  0.152 0.284
##                   lambda1 1.093 0.1585   0.540  0.822 1.452
##                   lambda2 1.544 0.2698   0.013  1.096 2.174
```

Selection plot: Beta model moderator

```r
selection_plot(sci_adj_ms_beta_mod)
```



Adjusted estimate in metaselection: Step 3PSM model moderator

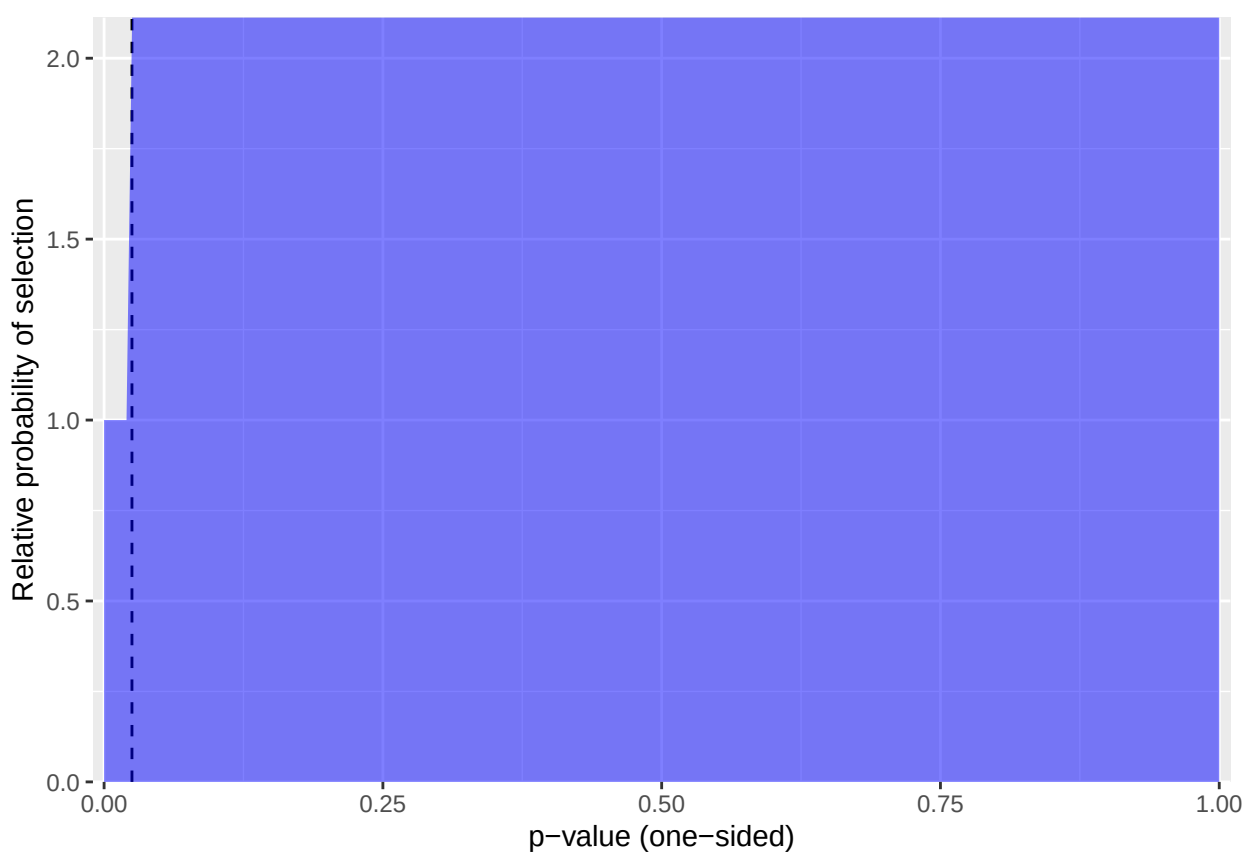
```
sci_adj_ms_step3_mod <- selection_model(
  dat = sci,
  yi = d,
  sei = sd_d,
  cluster = studyid,
  selection_type = "step",
  steps = 0.025,
  mean_mods = ~ outcome_type_author
)

sci_adj_ms_step3_mod
```

```
##                      param    Est     SE  p_value  CI_lo CI_hi
##          beta_(Intercept) 0.5014 0.1131 9.32e-06  0.280 0.723
##  beta_outcome_type_author 0.0961 0.1282 4.53e-01 -0.155 0.347
##                      tau2 0.1551 0.0326       NA  0.103 0.234
##                   lambda1 2.1122 0.5148 2.15e-03  1.310 3.406
```

Selection plot: Step 3 PSM model moderator

```
selection_plot(sci_adj_ms_step3_mod)
```



Adjusted estimate in metaselection: Step 4PSM model moderator

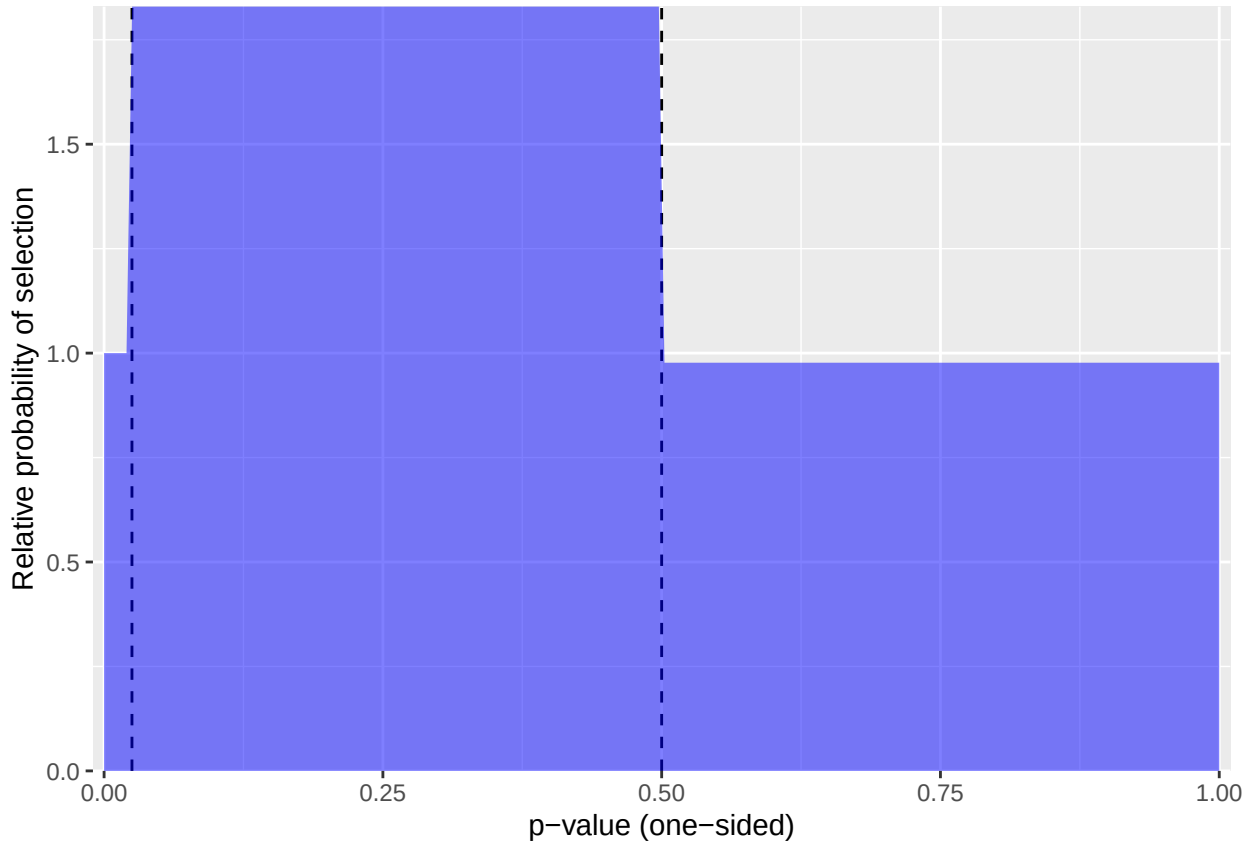
```
sci_adj_ms_step4_mod <- selection_model(
  dat = sci,
  yi = d,
  sei = sd_d,
  cluster = studyid,
  selection_type = "step",
  steps = c(0.025, .5),
  mean_mods = ~ outcome_type_author
)

sci_adj_ms_step4_mod
```

```
##                     param   Est     SE p_value  CI_lo CI_hi
##          beta_(Intercept) 0.386 0.1455 0.00793  0.101 0.671
##  beta_outcome_type_author 0.121 0.1525 0.42895 -0.178 0.420
##                      tau2 0.185 0.0343      NA  0.129 0.266
##                   lambda1 1.829 0.4219 0.00883  1.164 2.875
##                   lambda2 0.977 0.3609 0.94981  0.474 2.015
```

Selection plot: Step 4PSM model moderator

```
selection_plot(sci_adj_ms_step4_mod)
```


Relative probability of selection
1.5
1.0
0.5
0.0
0.00
0.25
0.50
0.75
1.00
p−value (one−sided)